Steven Tran

Senior Firmware Engineer

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SUMMARY

Results-driven Firmware Engineer with over a decade of experience in the full software development lifecycle for embedded systems. Specialized expertise in developing low-level firmware, bare-metal programming, and real-time operating systems (RTOS) for resource-constrained devices. Proven ability to architect and implement robust solutions using C/C++, optimize performance for power and memory, and integrate with hardware platforms at scale. Skilled in creating and maintaining board support packages (BSPs), developing device drivers, and collaborating with cross-functional teams to deliver production-grade firmware for consumer electronics and healthcare devices. Adept at driving innovation, adhering to best practices in firmware development and CI/CD, and aligning embedded strategies with business goals to deliver measurable impact.

SKILLS

Languages: C, C++, Assembly (ARM, x86), Python, Shell Script (Bash)

Microconterollers/Processores: ARM Cortex-M, ESP32, STM32, AVR, x86

Firmware/OS: Bare-Metal, FreeRTOS, Zephyr, Embedded Linux, Yocto Project

Hardware & Protocols: I2C, SPI, UART, USB, CAN, BLE, TCP/IP, Board Bring-up, Datasheets, Schematics, JTAG/SWD Debugging

Tools & CI/CD: Git, GDB, Oscilloscopes, Logic Analyzers, Jenkins, GitHub Actions, Docker, CMake, Make

Testing: Unity, CppUTest, Hardware-in-the-Loop (HIL) Testing, Unit Testing, Integration Testing, pytest

Architecture/Design Patterens: State Machines, Event-Driven Architecture, HAL (Hardware Abstraction Layer), Layered Architecture, Resource-Constrained Design

Cloud & DevOps:AWS (EC2, S3, Lambda, IoT Core), Azure IoT Hub, Google Cloud Platform (GCP), Databricks, Terraform, Kubernetes

Version Control: Git, GitHub, GitLab, Bitbucket

Collaboration & Productivity: Jira, Confluence, Slack, Postman, Swagger/OpenAPI

EXPERIENCE

Senior Firmware Engineer | Target | San Francisco, CA | 05/2024 – 08/2026

- Architected and developed embedded firmware in C/C++ for smart retail devices, focusing on low-power operation and real-time responsiveness..

- Implemented and optimized a board support package (BSP) and device drivers for new hardware platforms, reducing boot time by 35%.

- Developed and integrated firmware for BLE and Wi-Fi communication protocols, enabling seamless connectivity with mobile applications and cloud services.

- Created and maintained a Hardware Abstraction Layer (HAL) to ensure firmware portability across multiple microcontroller families, accelerating future product development.

- Implemented automated testing frameworks, including Hardware-in-the-Loop (HIL) tests, to monitor firmware stability and performance, reducing deployment errors by 25%.

- Utilized CI/CD pipelines with GitHub Actions and Jenkins to automate firmware builds, static analysis, and unit testing.

- Optimized firmware for memory and power consumption, extending battery life for portable devices by 20%.

- Participated in peer code reviews, ensuring adherence to best practices in embedded software development and MISRA C guidelines.

- Collaborated with hardware engineers and product managers to translate business objectives into technical firmware specifications with clear success metrics.

- Led efforts in automating firmware release and versioning workflows, reducing manual intervention by 40%.

- Maintained high standards of data privacy and security in compliance with corporate policies and healthcare regulations.

- Stayed current with emerging embedded systems trends and integrated novel low-power algorithms to enhance device capabilities and efficiency.

- Delivered technical presentations to stakeholders explaining complex hardware/software interactions in accessible terms, facilitating informed decision-making.

Senior Firmware Engineer | Walmart | Bentonville, Arkansas | 01/2020 – 04/2024

- Developed and deployed production-grade firmware for inventory management and logistics devices using C/C++, FreeRTOS, and Embedded Linux.

-Led the design and implementation of a sensor fusion pipeline to analyze real-time environmental data, improving inventory tracking accuracy by 18%.

- Worked extensively with low-level communication protocols (I2C, SPI, UART) to interface with a wide range of peripherals and sensors.

- Collaborated with DevOps teams to implement CI/CD pipelines for firmware deployment and monitoring using Jenkins, ArgoCD, and Docker.

- Optimized firmware for performance and computational efficiency, enhancing real-time data processing speed by 30%.

-Integrated device firmware with enterprise REST APIs and internal data platforms to support seamless operational workflows.

- Conducted performance monitoring and fine-tuning of RTOS tasks and memory management in collaboration with hardware teams.

- Provided mentorship and knowledge sharing sessions on best practices in firmware lifecycle management and reproducible builds.

- Implemented firmware governance strategies including version control, automated testing, and compliance adherence to ensure robustness and auditability.

- Partnered with business stakeholders to translate complex operational challenges into firmware-driven solutions with measurable KPIs.

- Automated hardware diagnostics and anomaly detection routines, improving device reliability by 22%.

- Contributed to the embedded systems strategy and roadmap development aligned with Walmart's digital transformation goals.

Firmware Engineer | Elevance Health | Indianapolis, Indiana | 09/2013 – 11/2019

- Developed early-stage embedded firmware for patient monitoring devices to predict health outcomes and optimize care pathways using C and Python.

- Built robust data acquisition and preprocessing pipelines for feature extraction from medical sensors, ensuring data integrity and HIPAA compliance.

- Collaborated with data scientists and clinical teams to define firmware requirements and translate healthcare challenges into embedded ML formulations.

- Implemented signal processing techniques for physiological data analysis to extract meaningful insights from sensor data.

- Supported firmware validation processes including unit testing, integration testing, and performance monitoring in staging environments.

- Worked with cloud infrastructure teams to deploy device data processing services on AWS EC2 and managed containerized microservices using Docker and Kubernetes.

- Applied optimization and mathematical programming techniques to resource allocation problems within healthcare operations.

- Contributed to documentation and peer reviews promoting software engineering best practices in firmware development.

- Engaged in continuous learning and experimentation with emerging embedded algorithms to enhance predictive analytics capabilities.

- Maintained adherence to HIPAA and corporate privacy policies ensuring member data confidentiality and security.

- Participated in cross-functional agile teams delivering iterative firmware solutions aligned with corporate mission and values.

- Supported integration of device firmware with enterprise health management systems and APIs for real-world application.

EDUCATION

Bachelor of Computer Science | University of Southern Mississippi, Hattiesburg, MS | 08/2009 – 05/2013